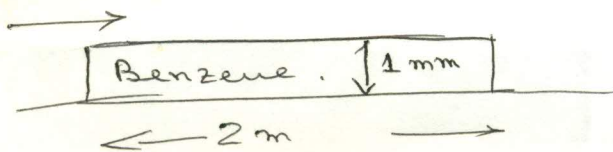


Benzene, a harmful chemical, has been spilled on the laboratory floor and has spread to a length of 2m. If a film 1 mm depth is formed, how long will it take for the Benzene to completely evaporate? Ventilation in the laboratory provides for airflow parallel to the surface at 1 m/s, and the benzene and air are both at 25 C. The mass densities of Benzene in the saturated vapor and liquid states are known to be 0.417 and 900 kg/m³, respectively. Given: For Air: ν (kinematic viscosity) = $15.7 \times 10^{-6} \text{ m}^2/\text{s}$; For Benzene-Air: $D_{AB} = 0.88 \times 10^{-5} \text{ m}^2/\text{s}$, $Sc = 1.78$.

Air 1m/s at 25°C.



$$P_{\text{sat.}} = 0.417 \text{ kg/m}^3$$

$$\rho_{\text{liq}} = 900 \text{ kg/m}^3$$

$$\nu_{\text{Air}} = 15.7 \times 10^{-6} \text{ m}^2/\text{s}, \quad D_{AB} = 0.88 \times 10^{-5} \text{ m}^2/\text{s} \quad Sc = 1.78$$

Calculate x_{tr} $Re_{tr} = 5 \times 10^5$

$$\frac{x_{tr} \cdot 1}{15.7 \times 10^{-6}} = 5 \times 10^5 \Rightarrow x_{tr} = 7.85 \text{ m.}$$

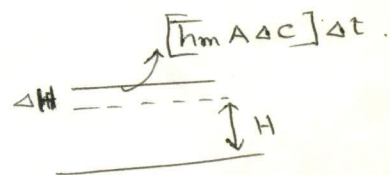
Thus the film will be under laminar flow

$$IN - OUT + acc = 0 \rightarrow -\dot{m} \Delta C \Delta t - A \Delta H \rho = 0.$$

Mass balance

$$-\frac{d}{dt}(PAH) = \dot{m}$$

$$= \bar{h}_m A \Delta C$$



$$\frac{dH}{dt} = -\frac{\bar{h}_m \Delta C}{\rho}$$

$$\int_H^0 dH = -\frac{\bar{h}_m \Delta C}{\rho} \int_0^t dt.$$

$$H = \frac{\bar{h}_m t \Delta C}{\rho} \Rightarrow \boxed{t = \frac{H \rho}{\bar{h}_m \Delta C}}$$

$h_m = ?$

$$Re_L = \frac{2 \times 1}{15.76 \times 10^{-6}} = 126903$$

$$\bar{Sh}_L = 0.664 Re_L^{1/2} Sc^{1/3} = 0.664 (126903)^{1/2} (1.78)^{1/3}$$

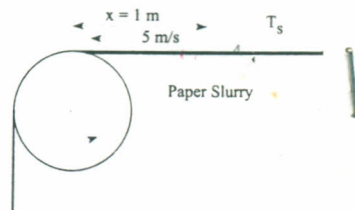
$$\frac{\bar{h}_m \times 2}{0.88 \times 10^{-5}} = 28606$$

$$\bar{h}_m = 1.26 \times 10^{-3}$$

$$t = \frac{H \rho}{\bar{h}_m \Delta C} = \frac{1 \times 10^{-3} \times 900}{1.26 \times 10^{-3} \times 0.417} \text{ s} = 1711 \text{ s. } (28.5 \text{ mins})$$

1. In a paper mill drying process, a sheet of paper slurry (water-fibre mixture) has a linear velocity of 5 m/s as it is rolled. Radiant heaters maintain a sheet temperature of $T_s = 330$ K, as evaporation occurs to dry, ambient air at 300 K, above and below the sheet. What is the evaporative flux at a distance of $x = 1$ m from the leading edge of the roll? What is the corresponding value of radiation flux (irradiation, G) that must be supplied to the sheet to maintain the temperature at 330 K? The sheet has an absorptivity of $\alpha = 1$.

Properties: Air ($T_f = 315$ K, 1 atm): $\gamma = 17.4 \times 10^{-6} \text{ m}^2/\text{s}$, $k = 0.0274 \text{ W/m.K}$, $Pr = 0.705$, Water vapor - Air ($T_f = 315$ K): $D_{AB} = 0.28 \times 10^{-4} \text{ m}^2/\text{s}$, $Sc = 0.616$, Saturated water vapor ($T_s = 330$ K): $\rho_{A, \text{sat}} = 0.1134 \text{ kg/m}^3$, $h_{fg} = 2366 \text{ kJ/kg}$



The drying process can be modeled as flow over a flat plate with heat and mass transfer.

For a unit area, at $x = 1$ m

$$\dot{m}''_{A, x} = h_{m, x} (p_{As} - p_{A\infty}) = h_{m, x} [p_{A, \text{sat}}(T_s) - \phi_{\infty} p_{A, \text{sat}}(T_a)]$$

$$\text{Evaluate } Re_x = \frac{U_{\infty} x}{\nu_B} = \frac{5 \text{ m/s} \times 1 \text{ m}}{17.40 \times 10^{-6} \text{ m}^2/\text{s}} = 2.874 \times 10^5$$

Since $Re_x < 5 \times 10^5 \Rightarrow$ laminar flow

$$Sh_x = \frac{h_{m, x} x}{D_{AB}} = 0.332 Re_x^{1/2} Sc^{1/3}$$

$$h_{m, x} = \frac{0.28 \times 10^{-4}}{1} \times 0.332 \times (2.874 \times 10^5)^{1/2} \times (0.616)^{1/3}$$

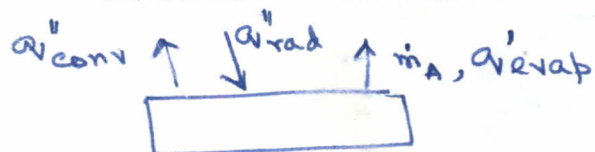
$$h_{m, x} = 4.24 \times 10^{-3} \text{ m/s}$$

Hence the evaporative flux at $x = 1$ m is

$$\dot{m}_A = 4.24 \times 10^{-3} \frac{\text{m}}{\text{s}} \times (0.1134 \frac{\text{kg}}{\text{m}^3} - 0)$$

$$\dot{m}_A = 4.81 \times 10^{-4} \frac{\text{kg}}{\text{s m}^2}$$

- b) From an energy balance on the differential element at $x = 1$ m.



$$q''_{\text{rad}} = q''_{\text{conv}} + q''_{\text{evap}}$$

$$q''_{\text{rad}} = h_x (T_s - T_a) + \dot{m}_A h_{fg}$$

Estimate h_x , use heat-mass transfer analogy

$$h_x = h_{m,x} \frac{k}{D_{AB}} \left(\frac{Pr}{Sc} \right)^{1/3}$$

$$= 4.24 \times 10^{-3} \frac{\text{m}}{\text{s}} \cdot \left(\frac{0.0274 \text{ W/mK}}{0.28 \times 10^{-4} \text{ m}^2/\text{s}} \right) \left(\frac{0.705}{0.616} \right)^{1/3}$$

$$h_x = 4.34 \text{ W/m}^2\text{K} \quad 1$$

Hence the radiant power required to maintain the slurry at $T_s = 330 \text{ K}$ is

$$q''_{\text{rad}} = 4.34 (330 - 300) + 4.81 \times 10^{-4} \times 2366 \times 10^3$$

$$q''_{\text{rad}} = (130 + 1138) \text{ W/m}^2 = 1268$$

$$q''_{\text{rad}} = 1268 \text{ W/m}^2 \quad 2$$

Consider a rotating disk suspended in an infinite fluid. One remarkable result for laminar flow situations was that the boundary layer thickness and hence transport coefficients were uniform across the surface of the disk. Using the 1/7th power law velocity profile, it has been shown that the boundary layer thickness in turbulent flow is a function of radial position:

$$\delta_h = 0.526 r \left(\frac{\gamma}{\omega r^2} \right)^{1/5} \quad \text{where} \quad Re = \frac{\omega r_o^2}{\gamma}$$

The analog of the friction factor, the average torque coefficient, $\overline{C_T}$, is defined by:

$$\overline{C_T} = \frac{2T}{\frac{1}{2} \rho \omega^2 r_o^5}$$

where r_o is the radius of the disk, T is the torque and ω is the rotation rate. In turbulent flow:

$$\overline{C_T} = 0.146 Re^{-1/5} \quad (1/7\text{th power law})$$

$$\frac{1}{\sqrt{\overline{C_T}}} = 1.97 \log_{10} (Re \sqrt{\overline{C_T}}) + 0.03 \quad (\text{Universal profile})$$

a) For a disk 10 cm in diameter, immersed in water at 25 °C, and spinning at 3000 rpm, what is the boundary layer thickness at the edge of the plate, and what is the total torque required to spin the disk?

b) The shear stress at the surface of the disk is proportional to the boundary layer thickness (δ_h), i.e.

$$\tau_w \approx \rho r \omega^2 \delta_h$$

Using the Chilton-Colburn analogy, determine the functional form of a correlation for the heat and mass transfer coefficient in turbulent flow. Given: kinematic viscosity, $\gamma = 8.58 \times 10^{-7} \text{ m}^2/\text{s}$

The first thing to do is to evaluate the Reynold's number, which can then be used to calculate the boundary layer thickness.

$$Re = \frac{\omega r_o^2}{\gamma} = \frac{3000 \times \frac{1}{60} \times (0.1)^2}{8.58 \times 10^{-7}} = 5.83 \times 10^5$$

$$\therefore \delta_h = 0.526 r_o \left(\frac{\gamma}{\omega r_o^2} \right)^{1/5} = 0.526 \times 0.05 \times \left(\frac{1}{5.83 \times 10^5} \right)^{1/5}$$

$$\delta_h = 1.85 \text{ mm.}$$

The torque can now be calculated

$$T = \frac{1}{4} \rho \omega^2 r_o^5 \overline{C_T}, \quad \overline{C_T} = 0.146 Re^{1/5} = 0.01$$

$$\therefore \frac{1}{\sqrt{\overline{C_T}}} = 1.97 \log_{10} (Re \sqrt{\overline{C_T}}) + 0.03 \Rightarrow \overline{C_T} = 0.324$$

Thus the torque is

$$T = \frac{1}{4} \rho \omega^2 r_o^5 \overline{C_T} = 1.95 \times 10^{-3} \text{ Nm} \quad (1/7\text{th Law})$$

$$T = \frac{1}{4} \rho \omega^2 r_o^5 \overline{C_T} = 6.31 \times 10^{-2} \text{ Nm} \quad (\text{Universal})$$

The Chilton-Coulburn analogy states that

$$\frac{\bar{C}_f Re}{2} = Nu Pr^{-1/3} = Sh Sc^{-1/3}.$$

where $\frac{\bar{C}_f Re}{2} = \frac{\tau_w Re}{\rho v^2}$

Thus

$$Nu = \frac{\rho \pi \omega^2 Sh Re}{\rho v^2} Pr^{1/3}$$
$$= \frac{\rho \pi \omega^2 (0.526 \pi Re^{-1/5}) Re}{\rho v^2} Pr^{1/3}$$

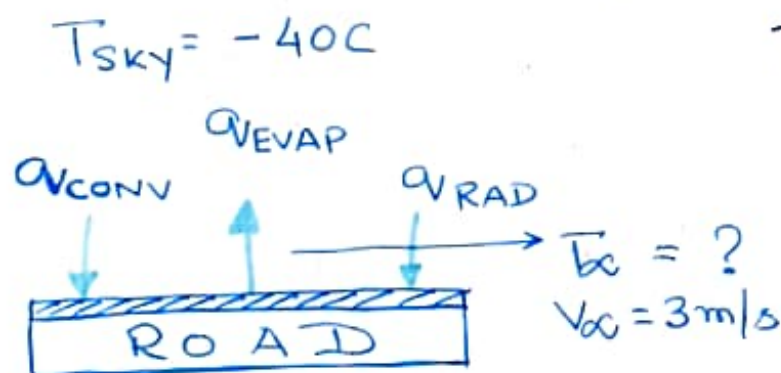
$$= \frac{0.526 \pi^2 \omega^2}{v^2} Re^{4/5} Pr^{1/3}.$$

and so

$$Sh = \frac{0.526 \pi^2 \omega^2}{v^2} Re^{4/5} Sc^{1/3}.$$

Some of the most dangerous road conditions exist on clear fall or spring nights after snow has thawed on the roadway. Although the air temperature may be above freezing, the water may freeze and form a thin layer of invisible ice, called black ice. Let us consider a roadway with a patch of water on its surface, 0.5m in length. The wind speed is 3 m/s, and the flow is laminar. The relative humidity of the ambient air is essentially zero. The roadway is made of asphalt is at 0°C and exposed to a night sky whose effective temperature is -40 °C. Additionally it can be assumed that the sky envelops the roadway such that the view factor between the road and the sky is 1. Assuming no water evaporates from the surface, at what air temperature will water just freeze? What would be the value of this temperature if evaporation is allowed? Properties: $\gamma_{aw} = 13.8 \times 10^{-6} \text{ m}^2/\text{s}$, $Pr = 0.705$, $k_{air} = 0.0243 \text{ W/m.K}$, $D_{aw} = 0.26 \times 10^{-5} \text{ m}^2/\text{s}$, $h_{fg} = 2502 \times 10^3 \text{ J/kg}$, $p^{sat} = 0.00611 \text{ atm}$, Emissivity (ϵ) = 0.85

The system can be modeled as a case of flow over a flat plate.



The three primary mechanisms for heat transfer (as shown) with the conservation equation

IN - OUT = ACCUM. (at SS)

$$Q_{CONV} - Q_{EVAP} - Q_{RAD} = 0 \quad \text{--- (1)}$$

$$Q_{CONV} = \bar{h} A_s (T_s - T_{\infty}), \quad Q_{RAD} = \sigma \epsilon F_{12} A_s (T_s^4 - T_{sky}^4)$$

$$Q_{EVAP} = \dot{m} h_{fg} = \bar{h}_m A_s (C_{as} - C_{\infty}) M_w h_{fg}$$

Here $F_{12} = 1$

$$Re = \frac{V_{\infty} L}{\nu_{air}} = \frac{3 \times 0.5}{13.8 \times 10^{-6}} = 108700 \text{ (LAMINAR)}$$

$$\bar{Nu} = 0.664 (Re)^{0.5} Pr^{1/3} = 195.7$$

$$\bar{h} = \frac{\bar{Nu} k_i}{L} = \frac{195.7 \times 0.0243}{0.5} = 9.51$$

$\text{W/m}^2 \text{K}$

For the case of NO EVAPORATION from EQ 1

$$\bar{h}(T_{\infty} - T_s) = \sigma \epsilon (T_s^4 - T_{sky}^4)$$

$$\therefore T_{\infty} = 273 + \frac{5.67 \times 10^{-8} \times 0.85 [(273)^4 - (233)^4]}{9.51}$$

$$T_{\infty} = 286.2 \text{ K} = 13.2 \text{ C}$$

(AIR T with NO evap. for black ice formation)

WITH EVAPORATION

If we consider evaporation, we can remove even more energy from the water and raise the air temperature for forming black ice still further.

$$\overline{Sh}_L = 0.664 (Re_L)^{1/2} Sc^{1/3} = 177.2$$

$$\bar{h}_m = 9.22 \times 10^{-6} \text{ m/s}$$

$$\frac{\dot{m}}{A_s} = \bar{h}_m M_w \left(\frac{p_{sat}}{RT} - \frac{p_{\infty}}{RT} \right)$$

= 0.

$$= \frac{9.22 \times 10^{-6} \times 18 \times 0.006 \times 100}{8.31 \times 10^{-2} \times 273}$$

$$= 4.39 \times 10^{-5} \text{ kg/m}^2 \text{ s}$$

From eq. (i)

$$T_{\infty} = \frac{\sigma \epsilon (T_s^4 - T_{sky}^4) + \frac{\dot{m} h_{fg}}{h_{AS}}}{h} + T_s$$

$$T_{\infty} = 25.2 \text{ C.}$$

Therefore evaporation does increase the air T that can be sustained while still allowing the water to freeze.

These T's are a bit too high. The error may have come from neglecting the conduction from the soil.